Stats 20, Project 2 – Spring 25

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06/01/26

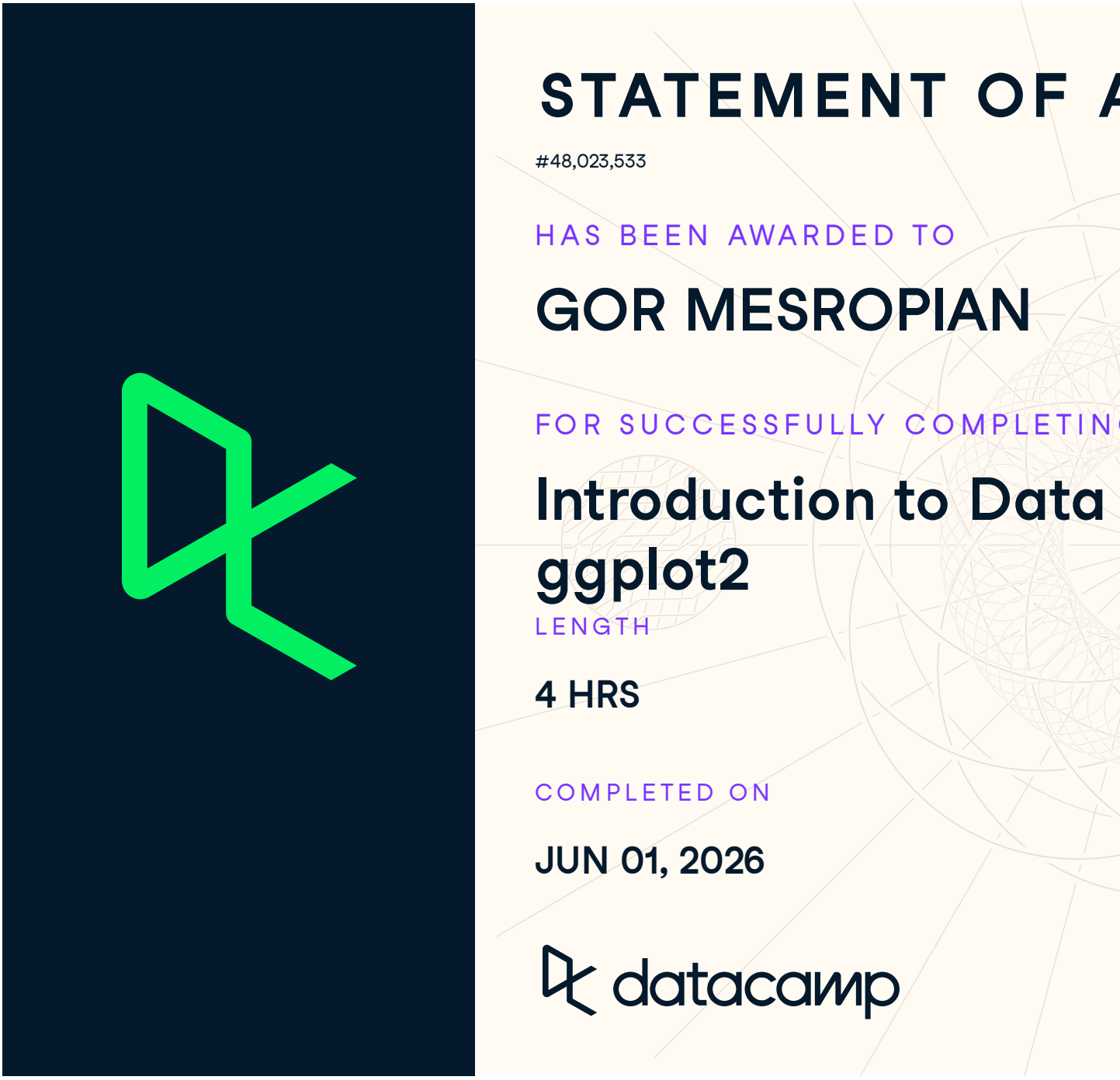
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Part 1: The Campening

(Paste Image of Certificate Below)

```
knitr::include_graphics("certificate_project2.jpg")
```



Part 2: The Penguining

(a)

```
#install.packages("palmerpenguins")
library(palmerpenguins)
```

```
##
```

```
## Attaching package: 'palmerpenguins'

## The following objects are masked from 'package:datasets':
##
##     penguins, penguins_raw
```

```
data(penguins)
```

(b)

```
head(penguins)
```

```
## # A tibble: 6 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           NA            NA                  NA          NA
## 5 Adelie  Torgersen           36.7          19.3               193        3450
## 6 Adelie  Torgersen           39.3          20.6               190        3650
## # i 2 more variables: sex <fct>, year <int>
```

```
str(penguins)
```

```
## tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
##  $ species          : Factor w/ 3 levels "Adelie","Chinstrap",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ island           : Factor w/ 3 levels "Biscoe","Dream",..: 3 3 3 3 3 3 3 3 3 3 ...
##  $ bill_length_mm   : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
##  $ bill_depth_mm    : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
##  $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
##  $ body_mass_g      : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
##  $ sex              : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
##  $ year             : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

```
summary(penguins)
```

```
##       species          island    bill_length_mm  bill_depth_mm
##  Adelie   :152   Biscoe   :168   Min.   :32.10   Min.   :13.10
##  Chinstrap: 68   Dream    :124   1st Qu.:39.23   1st Qu.:15.60
##  Gentoo   :124   Torgersen: 52   Median :44.45   Median :17.30
##                                  Mean   :43.92   Mean   :17.15
##                                  3rd Qu.:48.50   3rd Qu.:18.70
##                                  Max.   :59.60   Max.   :21.50
##                                  NA's   :2       NA's   :2
##  flipper_length_mm  body_mass_g       sex           year
##  Min.   :172.0     Min.   :2700   female:165   Min.   :2007
##  1st Qu.:190.0     1st Qu.:3550   male  :168   1st Qu.:2007
##  Median :197.0     Median :4050   NA's  : 11   Median :2008
```

```
##  Mean   :200.9     Mean   :4202                Mean   :2008
##  3rd Qu.:213.0     3rd Qu.:4750                3rd Qu.:2009
##  Max.   :231.0     Max.   :6300                Max.   :2009
##  NA's   :2         NA's   :2
```

(c)

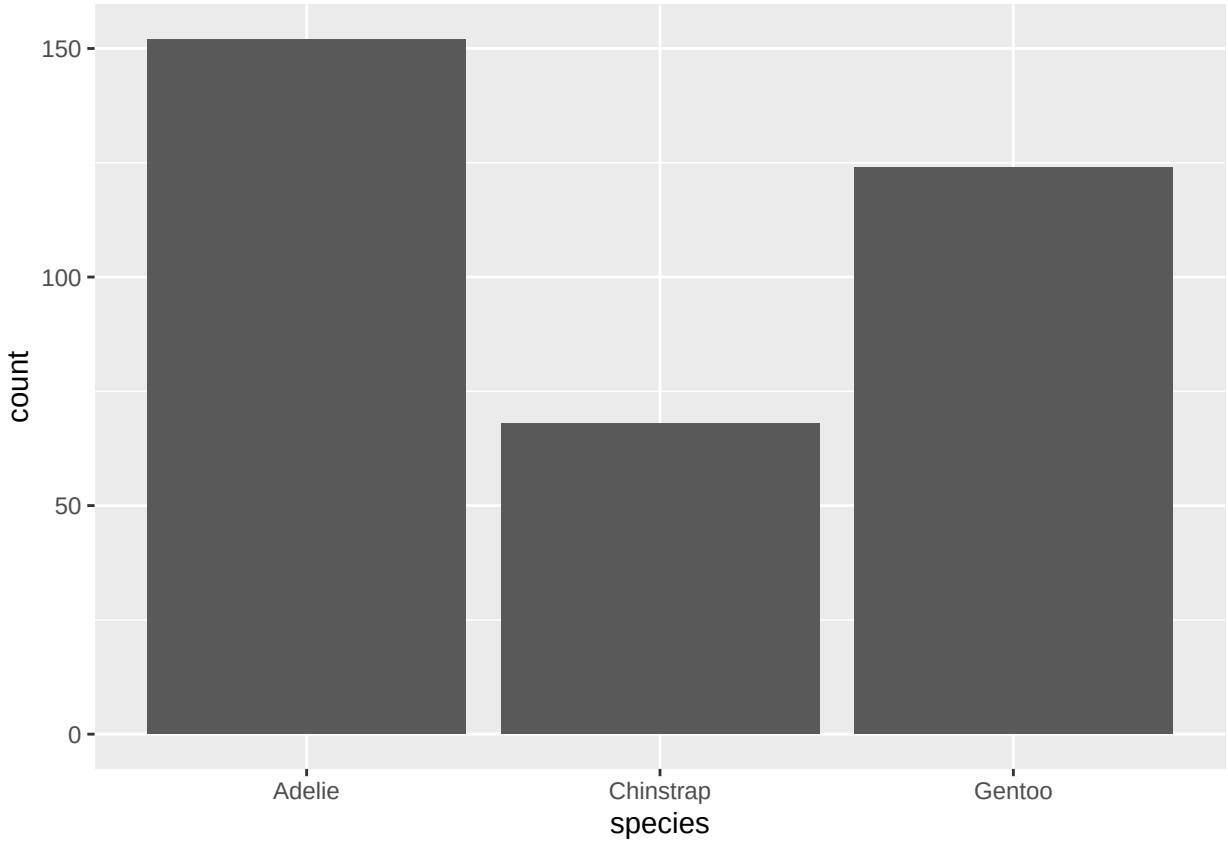
```
penguins2 <- read.table("penguins2.txt", header = FALSE)
penguins2 <- penguins2[, -1] #removing first column
colnames(penguins2) <- colnames(penguins)


#using this command to check if they are the same
all.equal(penguins, tibble::as_tibble(penguins2))
```

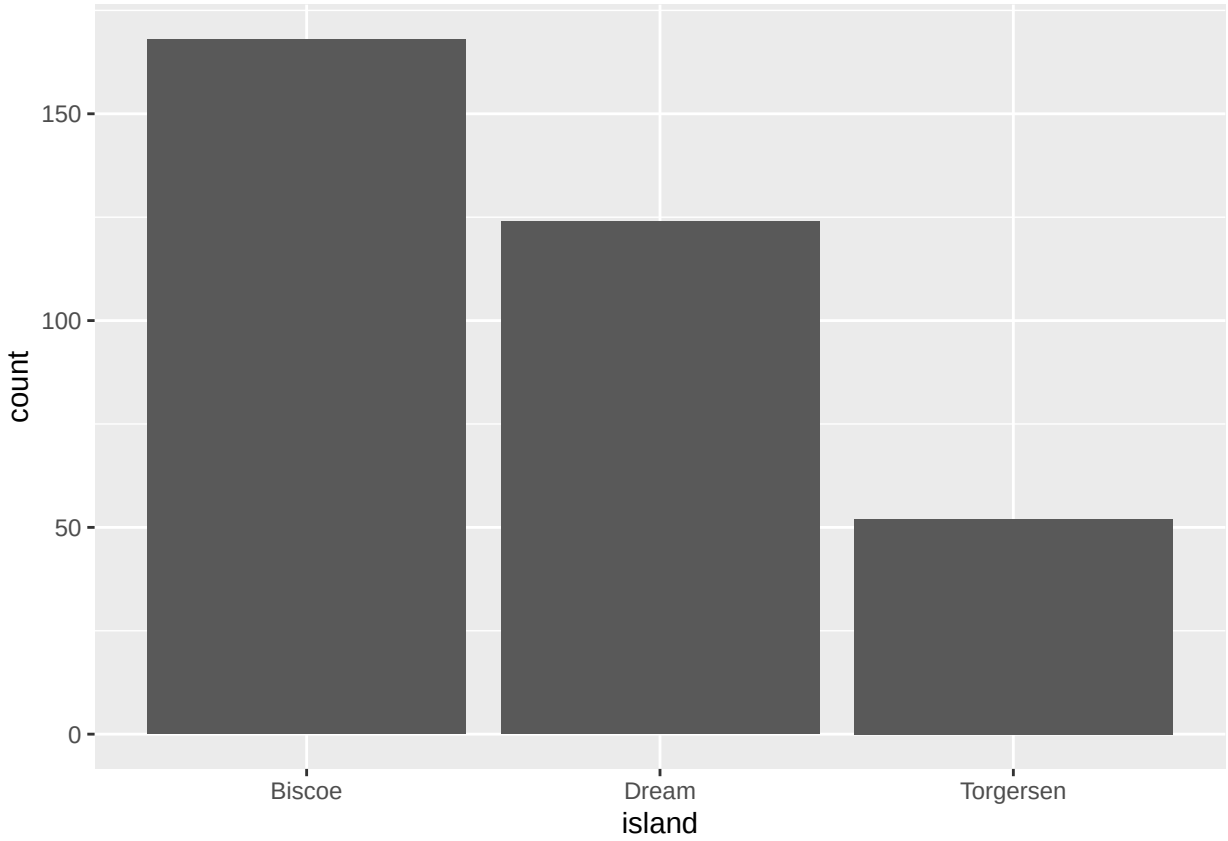
```
## [1] "Component \"species\": 'current' is not a factor"
## [2] "Component \"island\": 'current' is not a factor"
## [3] "Component \"sex\": 'current' is not a factor"
```

(d)

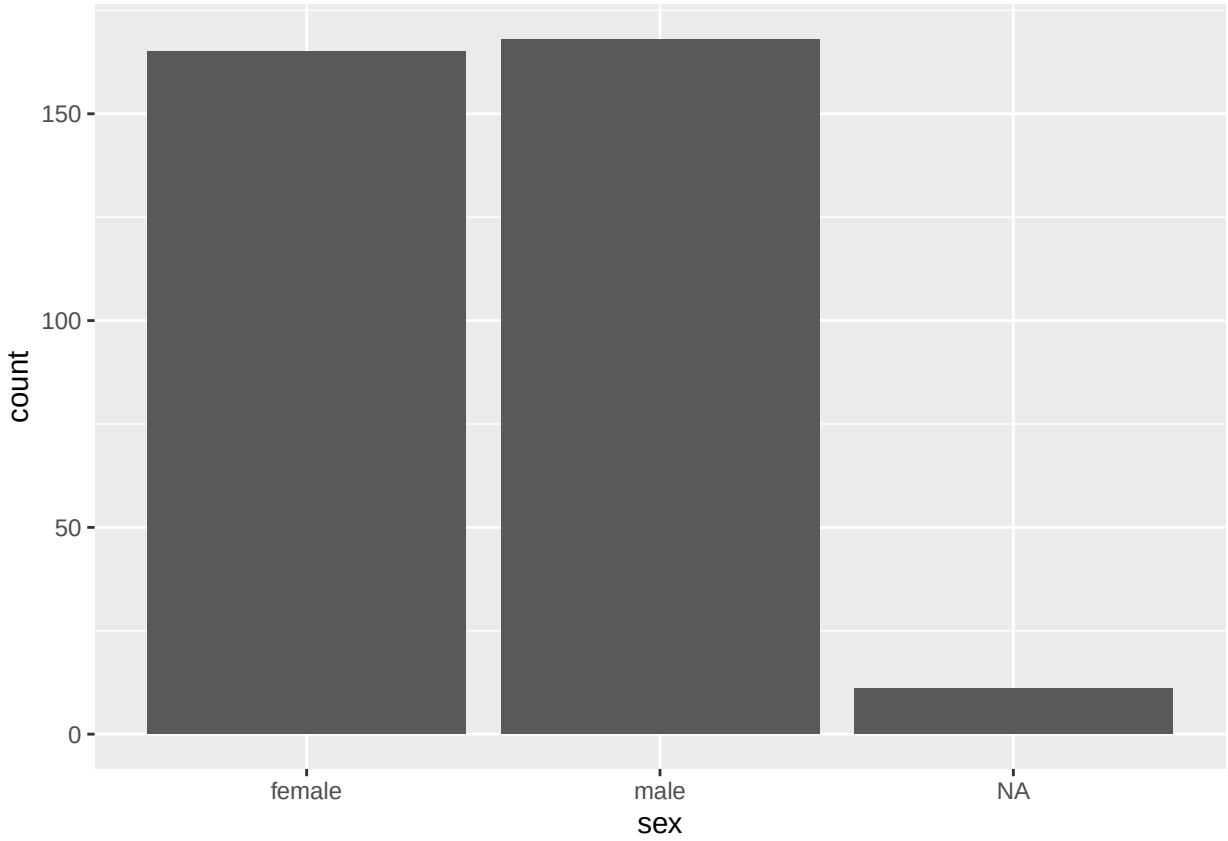
```
library(ggplot2)
#use geom bar for categorical variables and histogram for continuos data variables
# Species
ggplot(penguins, aes(x = species)) + geom_bar()
```



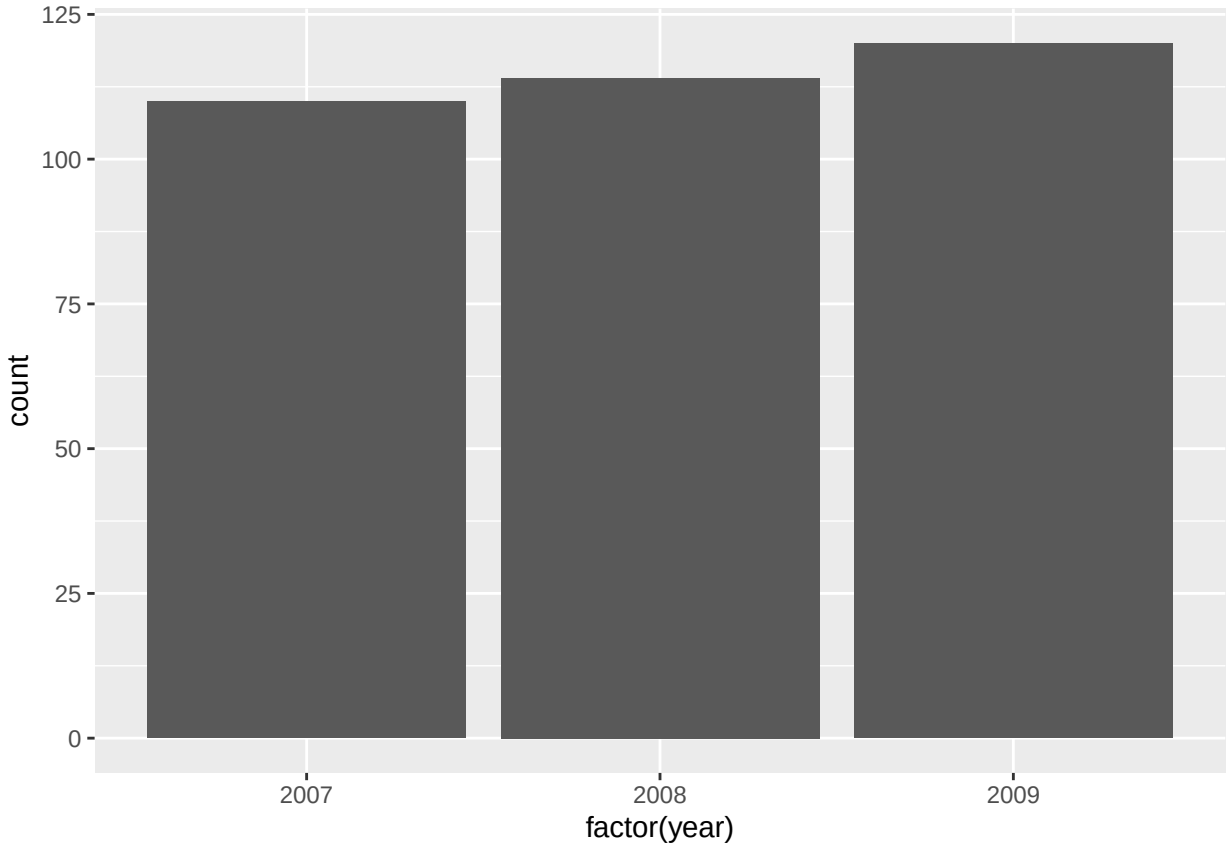

```
# Island
ggplot(penguins, aes(x = island)) + geom_bar()
```




```
# Sex
ggplot(penguins, aes(x = sex)) + geom_bar()
```



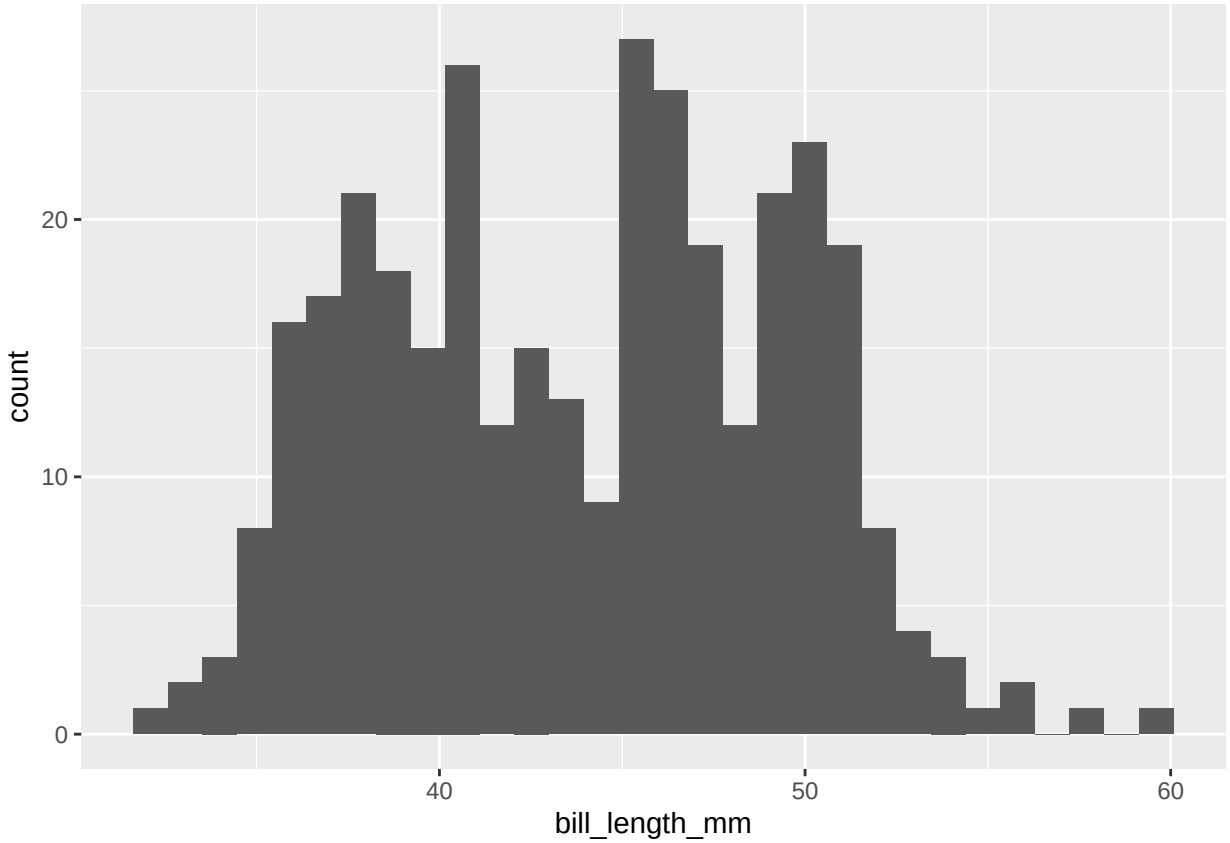

```
# Year
ggplot(penguins, aes(x = factor(year))) + geom_bar()
```




```
# Bill length
ggplot(penguins, aes(x = bill_length_mm)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

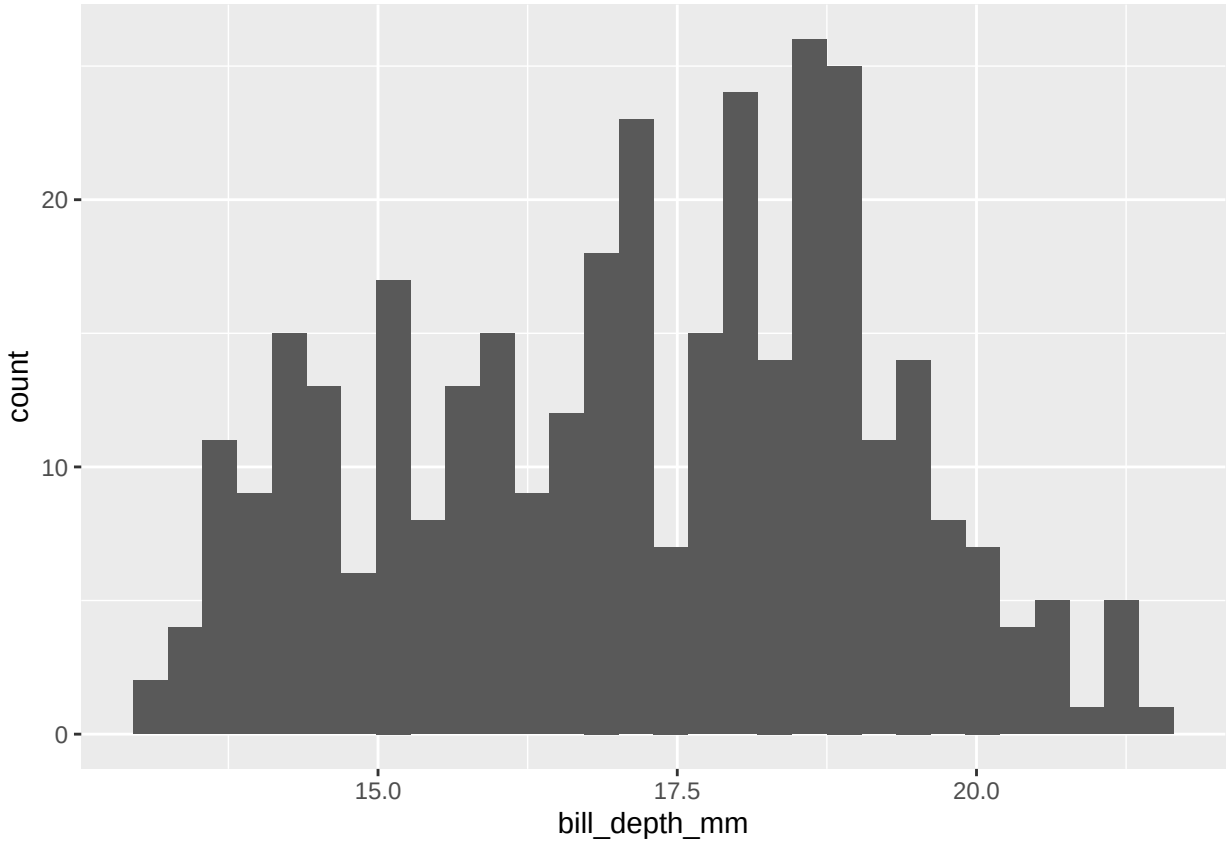
```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
# Bill depth
ggplot(penguins, aes(x = bill_depth_mm)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

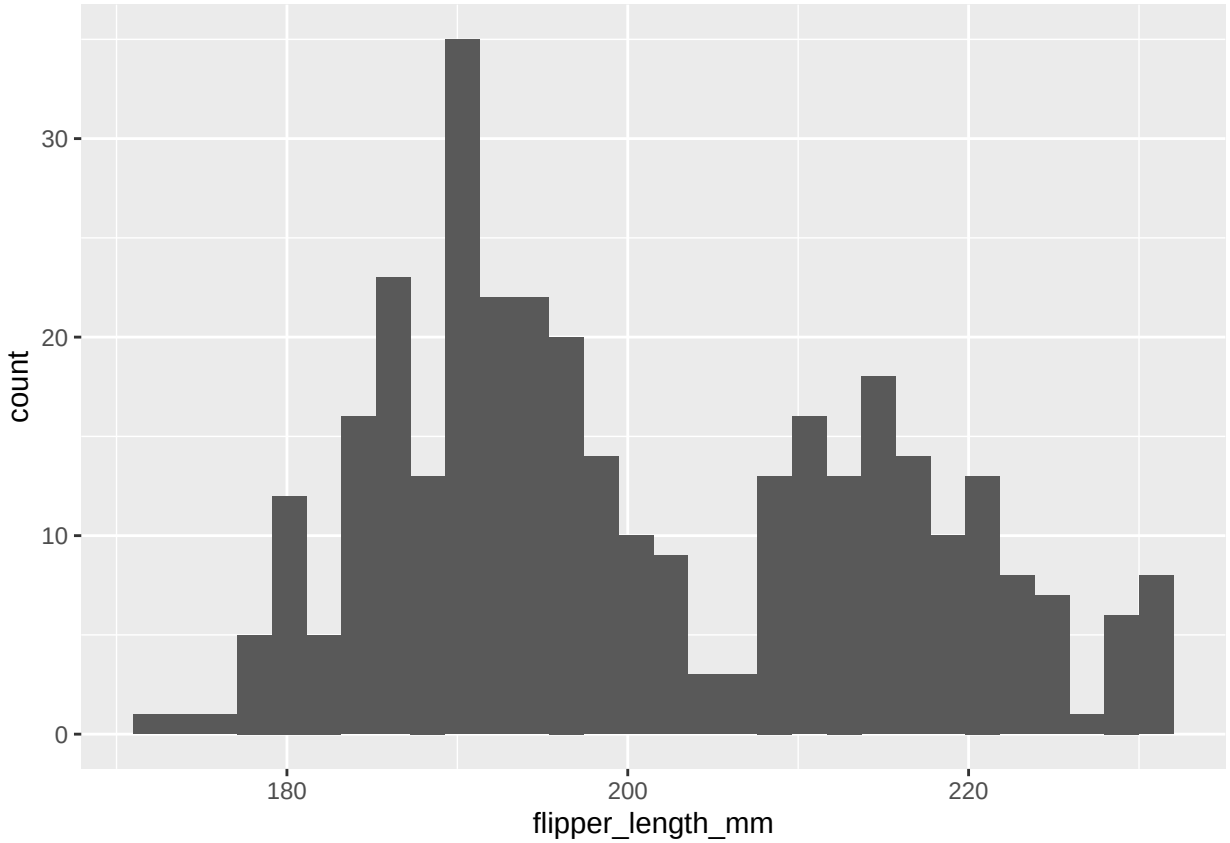
```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
# Flipper length
ggplot(penguins, aes(x = flipper_length_mm)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

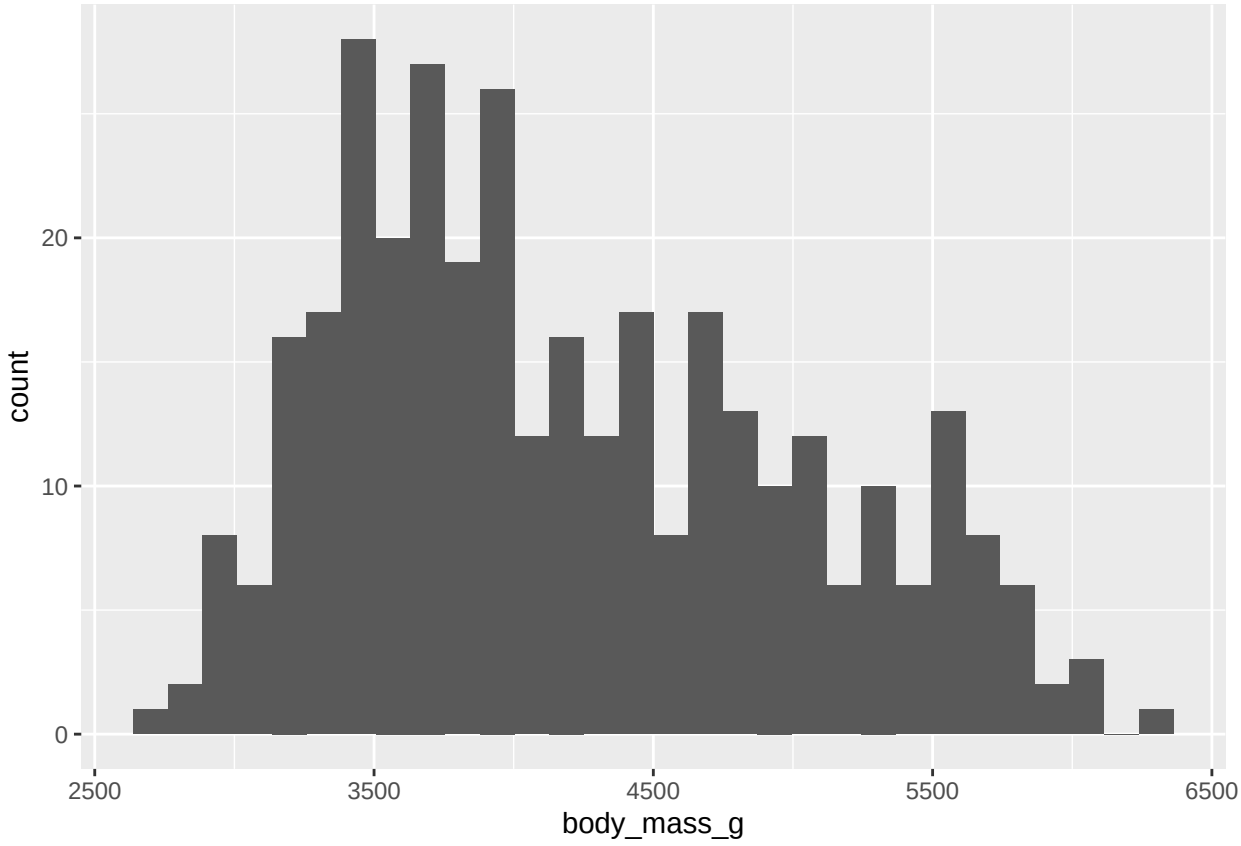
```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
# Body mass
ggplot(penguins, aes(x = body_mass_g)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

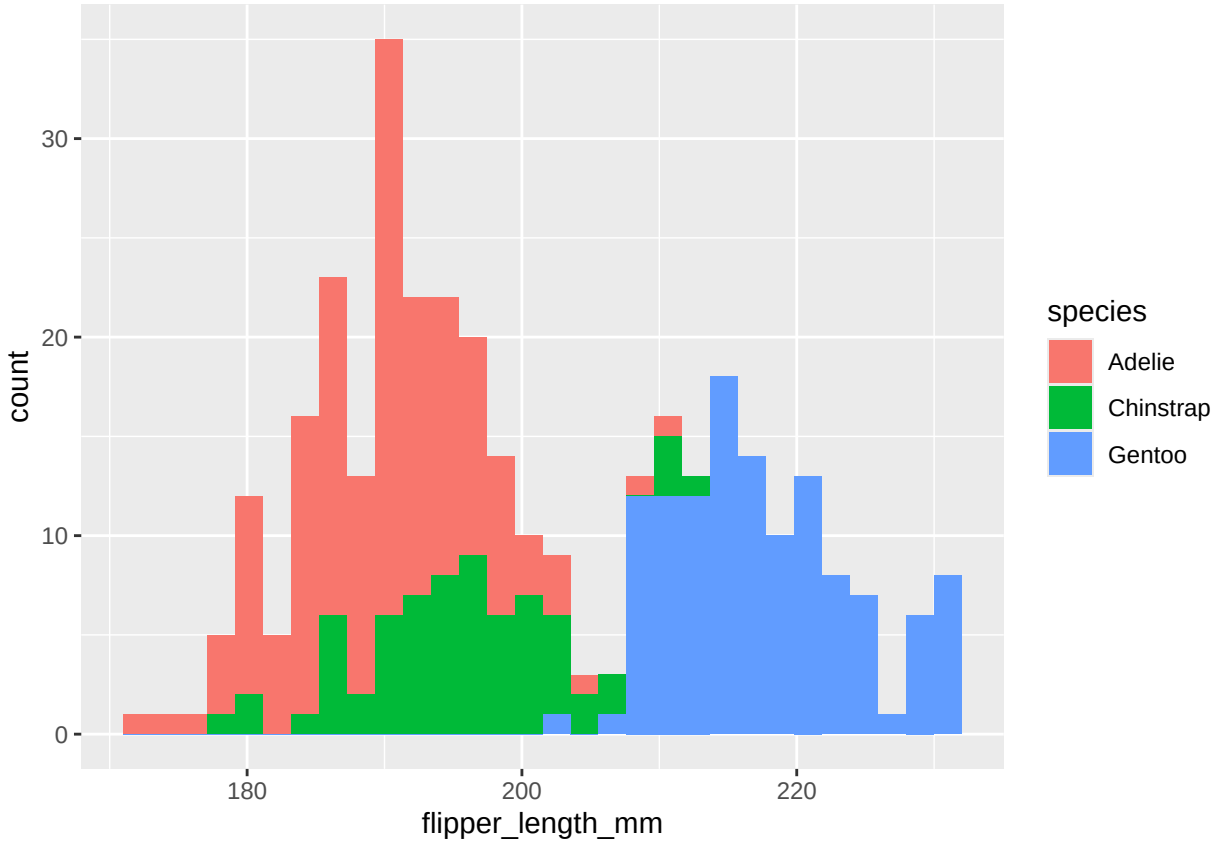


(e)

```
ggplot(penguins, aes(x = flipper_length_mm, fill = species)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

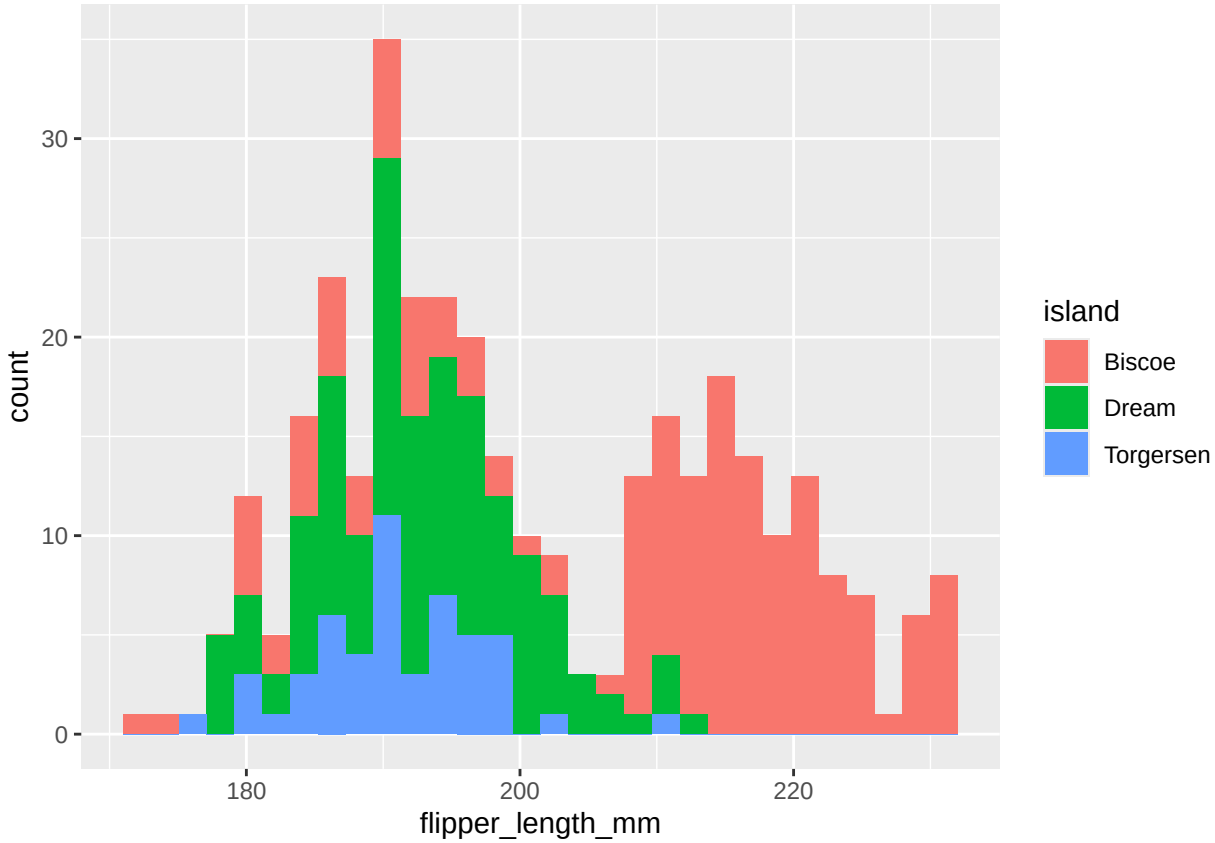
```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(penguins, aes(x = flipper_length_mm, fill = island)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.

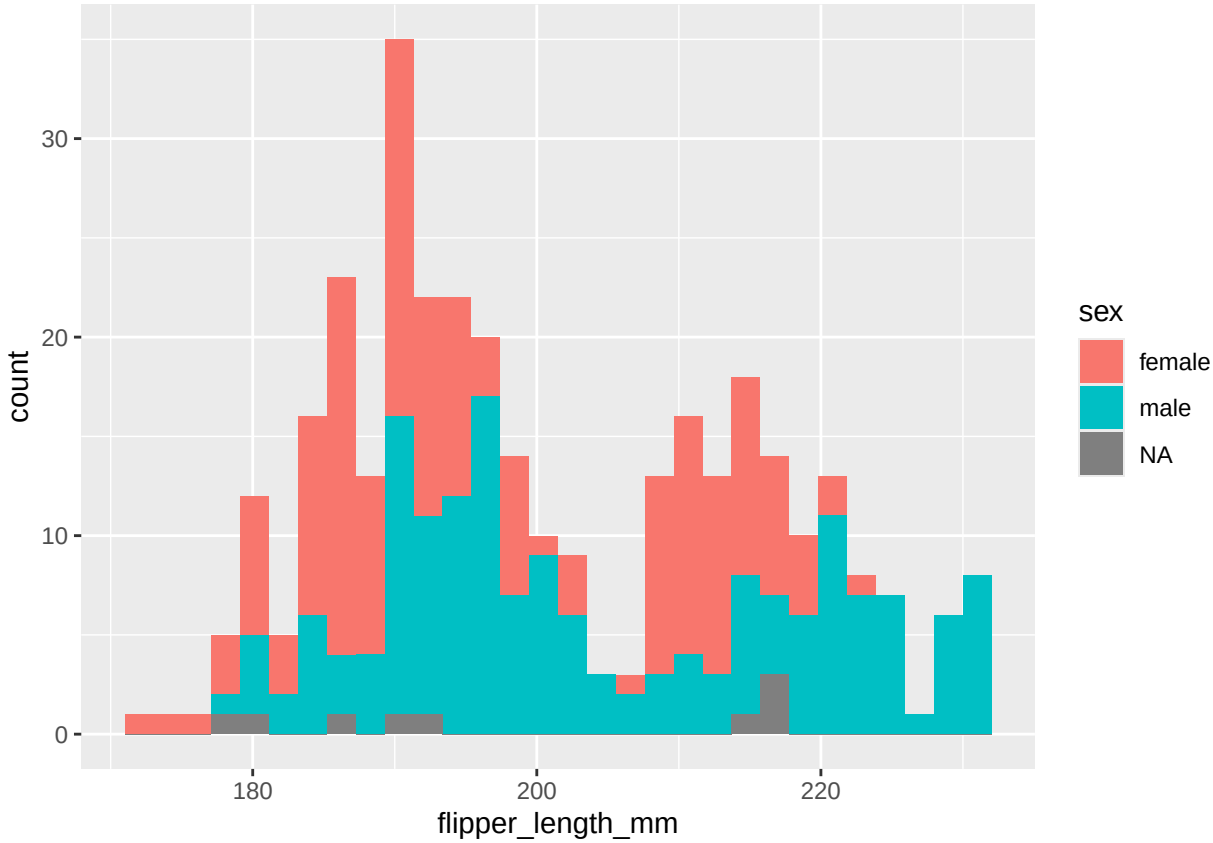
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(penguins, aes(x = flipper_length_mm, fill = sex)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

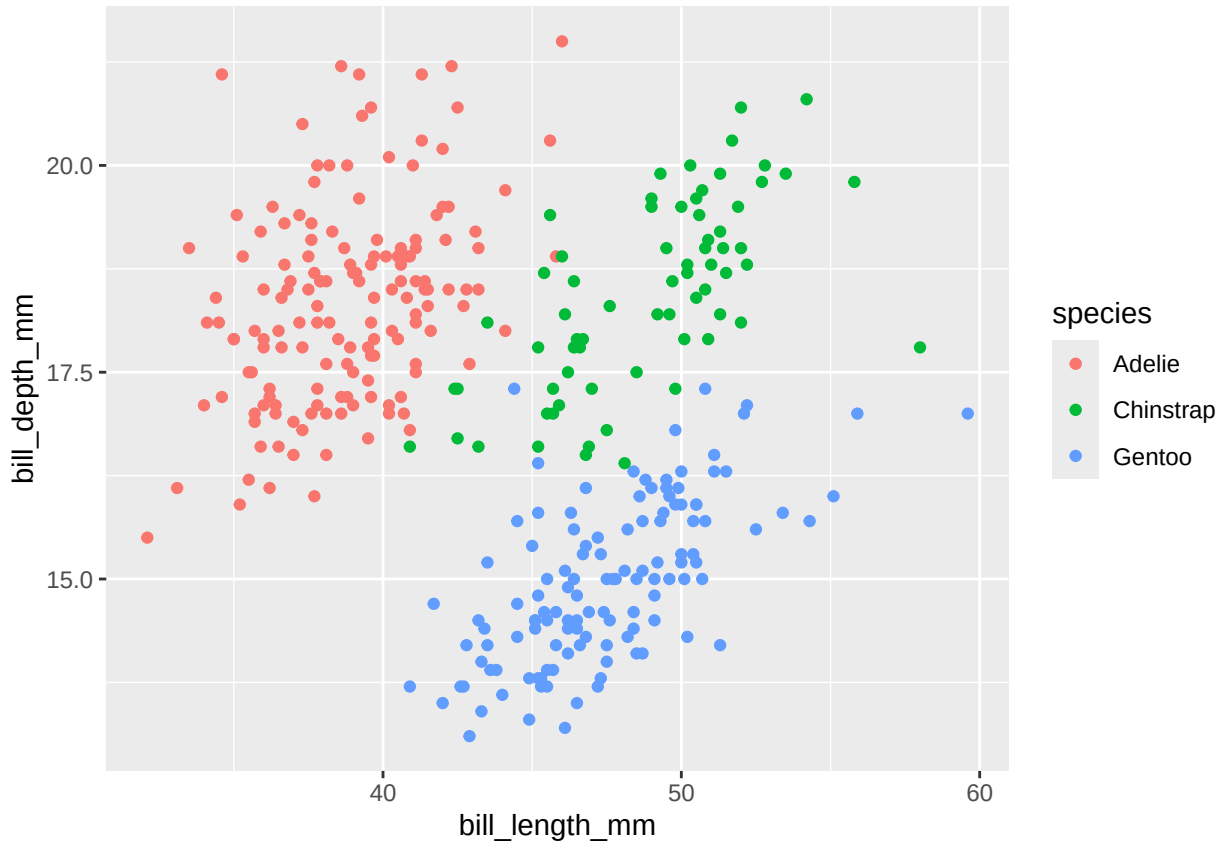



```
#the presence of different species explains why the data for flipper length is bimodal, to find this ou
```

(f)

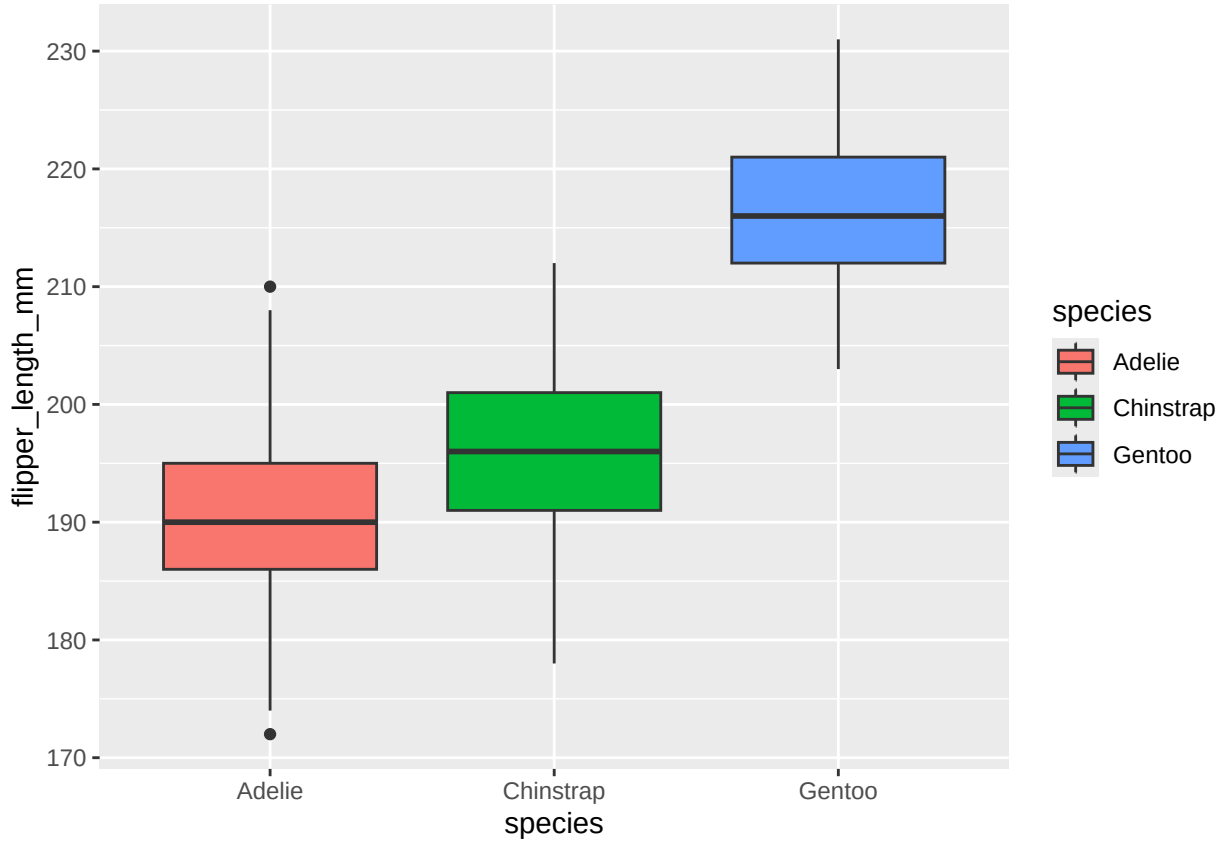
```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm, color = species)) +
  geom_point() #for bill length and depth, Adelie has the shortest but deep bill, chinstrap has a longe
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



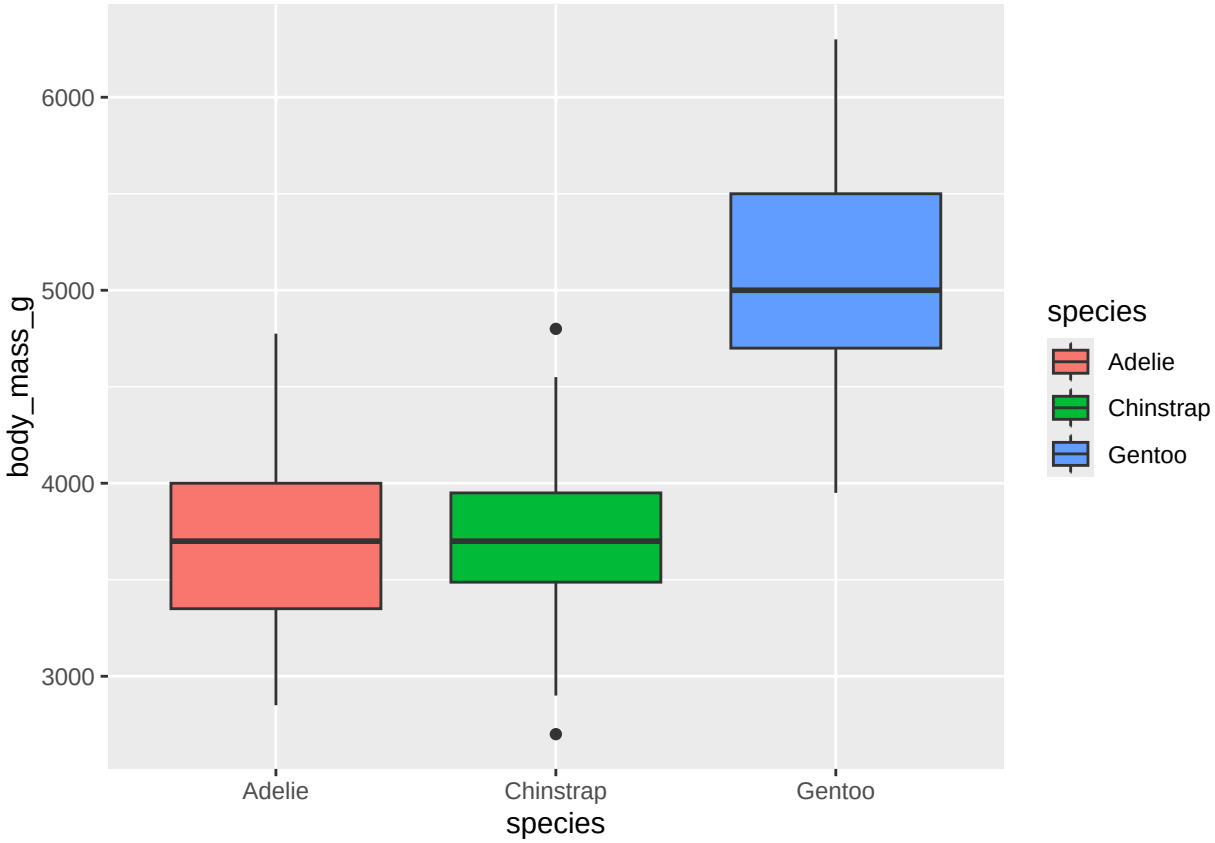

```
# flipper length
ggplot(penguins, aes(x = species, y = flipper_length_mm, fill = species)) +
  geom_boxplot()
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
#for flipper length, Gentoo has the largest by far, then chinstrap is on average sligtly larger than Ad
# body mass
ggplot(penguins, aes(x = species, y = body_mass_g, fill = species)) +
  geom_boxplot()
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

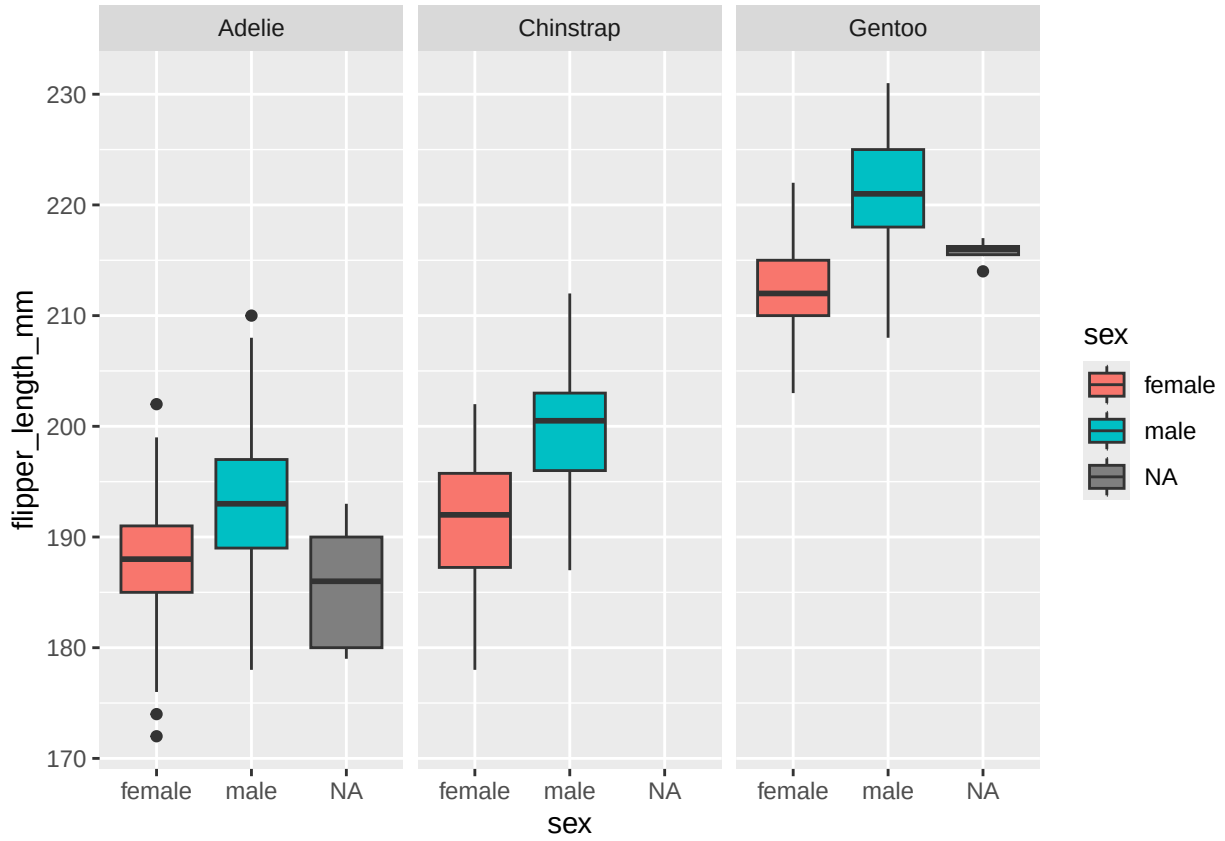



```
#for body mass, Gentoo is again the highest, Chinstrap and Adelie has similar mass but Adelie penguins
```

(g)

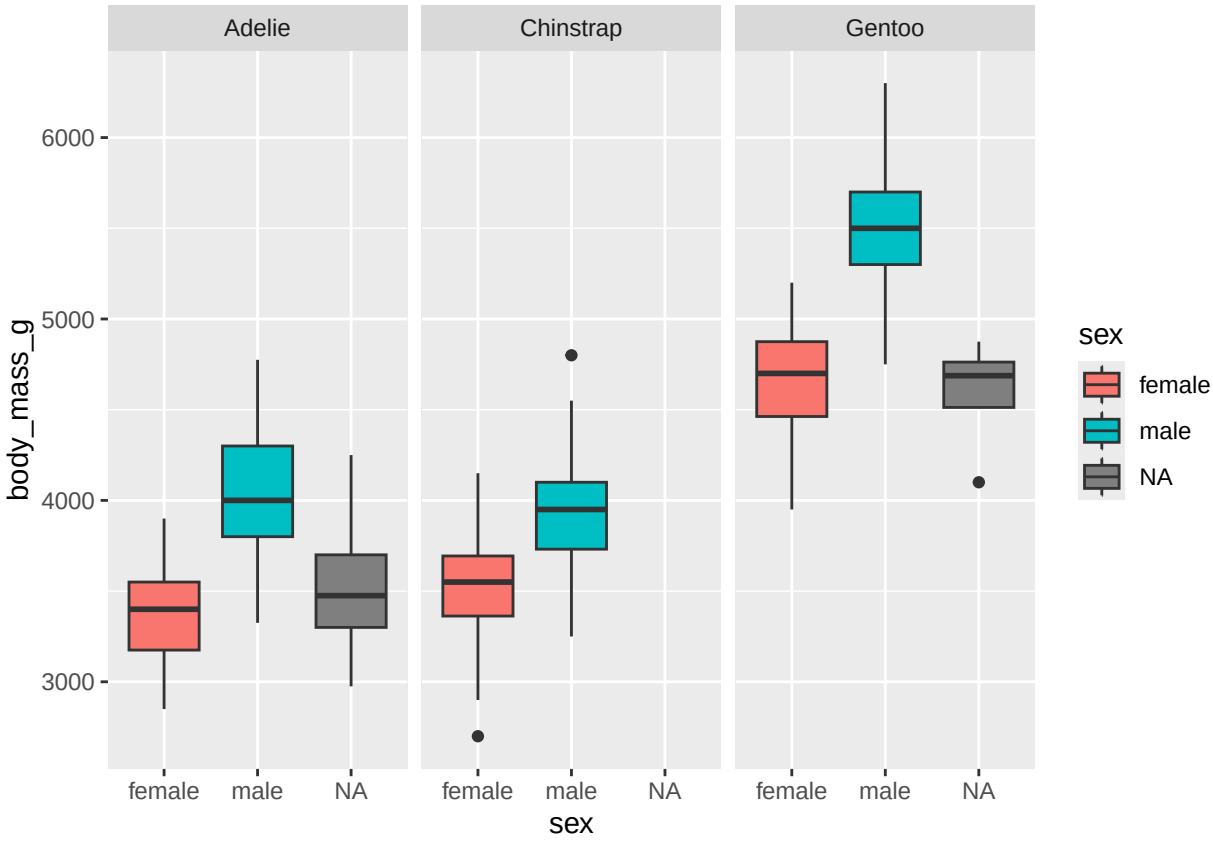
```
ggplot(penguins, aes(x = sex, y = flipper_length_mm, fill = sex)) +
  geom_boxplot() +
  facet_wrap(~ species)
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
ggplot(penguins, aes(x = sex, y = body_mass_g, fill = sex)) +
  geom_boxplot() +
  facet_wrap(~ species)
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```



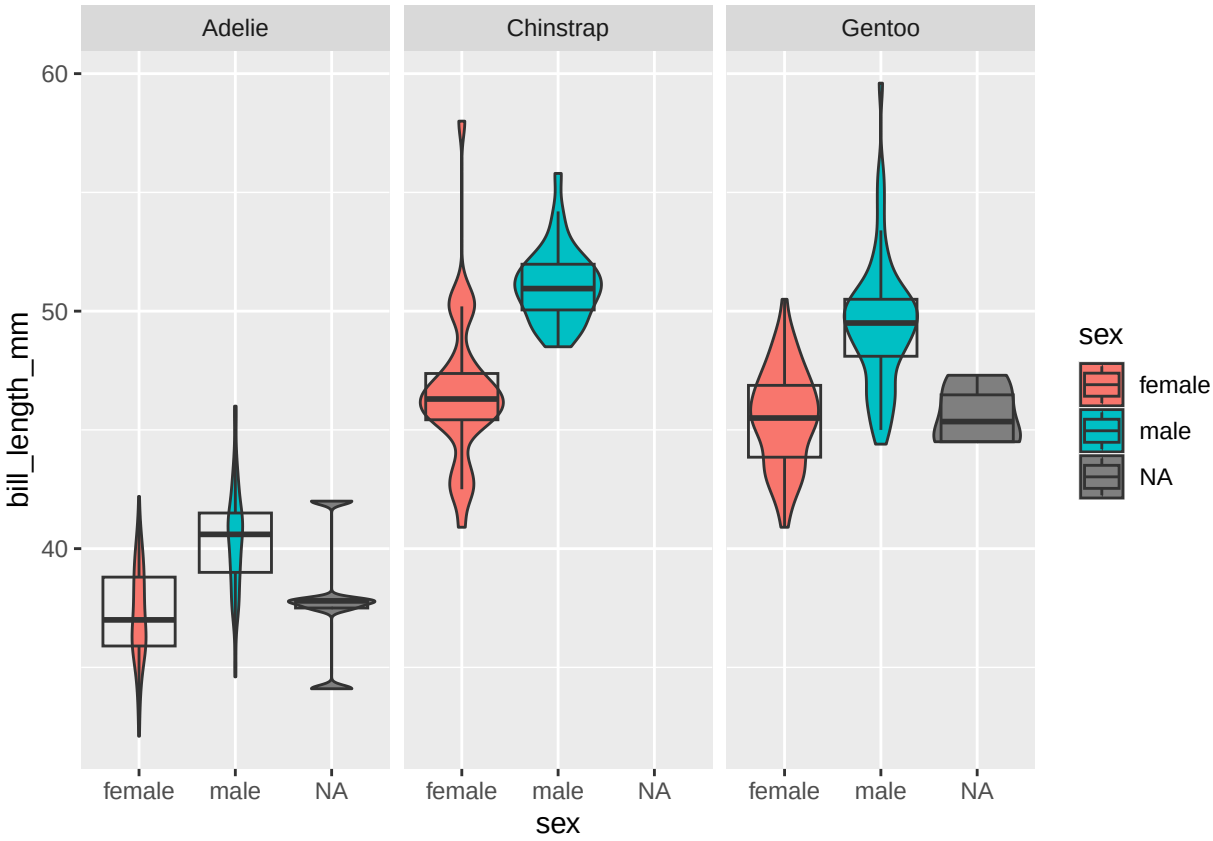

```
# Within every species, males are consistently heavier and have longer flippers than females
```

(h)

```
ggplot(penguins, aes(x = sex, y = bill_length_mm, fill = sex)) +
  geom_violin() +
  geom_boxplot(alpha = 0) +
  facet_wrap(~ species)
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_ydensity()`).
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
#A violon plot shows us the shape of the distributions better but its harder to read the summary statis

#Boxplos are good at showing median, spread, and any outlisers but they do not show the distributions
```

(i)

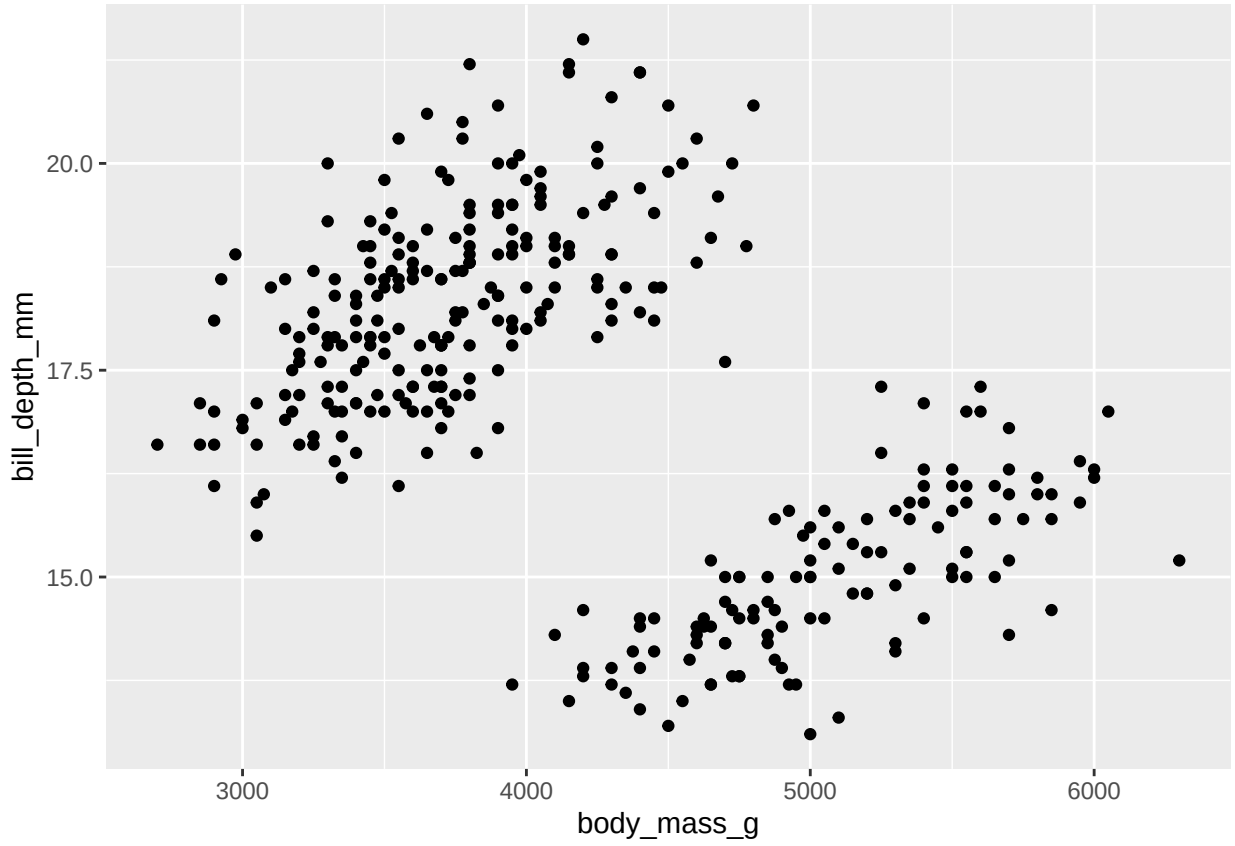
```
cor(penguins$body_mass_g, penguins$bill_depth_mm, use = "complete.obs")
```

```
## [1] -0.4719156
```

```
#returns a -0.47 correlation which means that larger penguins will tend to have less deep bills. Honest


#build plot to show relationship
ggplot(penguins, aes(x = body_mass_g, y = bill_depth_mm)) +
  geom_point()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
#in reality the relationship is a positive orreleation however within each species, when we lump all th
```

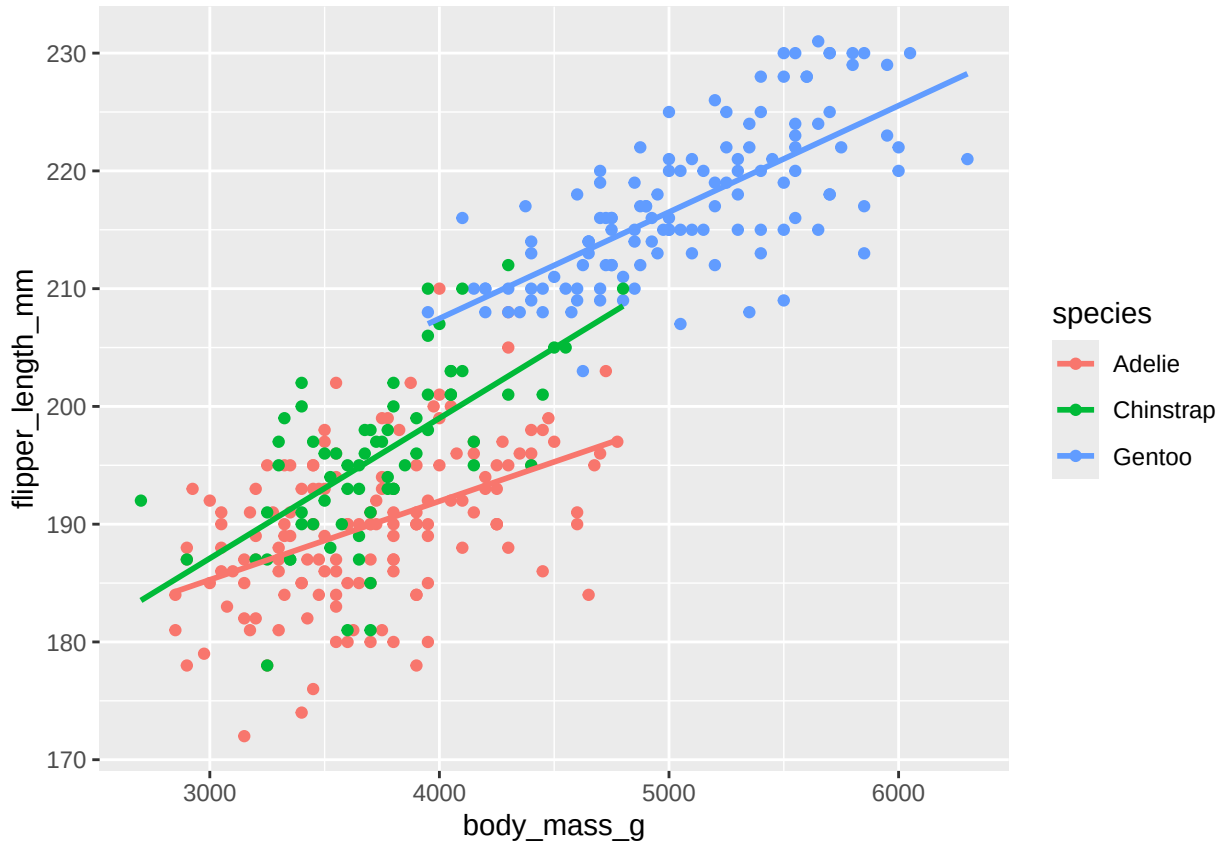
(j)

```
# TEST: Is flipper length correlated with body mass?
ggplot(penguins, aes(x = body_mass_g, y = flipper_length_mm, color = species)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
#I want to test for the p value here. to be true we are looking for a value <0.05
cor.test(penguins$body_mass_g, penguins$flipper_length_mm, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  penguins$body_mass_g and penguins$flipper_length_mm
## t = 32.722, df = 340, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.843041 0.894599
## sample estimates:
##       cor
## 0.8712018
```

```
#p-value < 2.2e-16 This is really small therefore we accept that flipper length is correlated with body
```